INCH-POUND

MIT-14442D(AT)
28 June 1990
SUPERSEDING
MIL-T-14442C(AT)
20 September 1967

MILITARY SPECIFICATION

TANK, FUEL, ENGINE: NON-METALLIC, NON-SELF-SEALING, FOR MILITARY VEHICLES

This specification is approved for use within the U.S. Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of non-metallic, inn-self-sealing fuel tanks (fuel cell type) for use in military tactical vehicles and combat vehicles. Configuration of fuel tanks shall be in accordance with applicable drawings (see 3.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 **Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2910

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SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-S-735	- Standard Test Fluids; Hydrocarbon.
VV-F-800	- Fuel Oil, Diesel.
PPP-T-76	- Tape, Packaging, Paper (for Carton Sealing].

MILITARY

MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys .
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal services .

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Navy Publication and Printing Service Office, Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue are those cited in the solicitation.

DRAWINGS

ARMY

7993036	- Gas Tank Cell, Right.
7993037	- Gas Tank Call, Left.
8340664	- Fuel Cell Assembly.
10918604	- Tank, Fuel, Engine.
10952618	- Cell Assembly, Fuel, Engine (Center).
11633639	- Tank, Fuel, Engine.
11647693	- Bladder, Fuel.

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(Copies of drawings and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 **Non-Government publications.** The following documents form apart of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D361	Existent Gum in Fuels by Jet Evaporation, Test Method for.
ASTM D412	Rubber Properties in Tension, Test Methods for.
ASTM D413	Rubber Property-Adhesion to Flexible Substrate, Test Methods for.
ASTM B633	Standard Specification for Electrodeposited Coating of Zinc on Iron and Steel.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organization which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets, MS standards, or drawings), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.4).

3.2 Materials. Materials used in the fabrication of fuel tanks shall be in accordance with-applicable specifications or drawings (see 4.7.1 and 6.4)

3.2.1 **Recycled, virgin and reclaimed materials.** There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.7.1 and 6-5.2).

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3.2.2 Protective treatments.

3.2.2.1 Fittings. Steel fittings, except stainless, when furnished and where paint is not required, shall be zinc plated in accordance with ASTM B633, Type II, class Fe/Zn 5. Steel fittings, when furnished and which are to be painted, shall be cadmium plated in accordance with type III, class I of QQ-P-416. Aluminum fittings shall be anodized in accordance with MIL-A-8625 (see 4.7.1).

3.2.2.2 Rubber molded fittings.All openings to tanks shall be fitted with fully cured Buns-N or equally fuel resistant material which shall be resistant to cutting, and to gouging as a result of sealing pressure generated by torquing hold-down bolts (see 4.7.1).

3.3 Design and construction. The fuel tank shall be of flexible construction, and shall consist generally of a nonmetallic carcass of unit construction, with openings or fittings satisfactory for the necessary attachment of parts and accessories in accordance with applicable drawings. Dimensions and contour shall be as specified on applicable drawings (see 4.7.1, 4.7.2, and 6.2). The following is a partial listing of engine fuel tank drawings that make reference to this specification:

Drawing number	Title	Vehicle applications
7993036	Gas Tank Cell, Right	M53, M55
7993037	Gas Tank Cell, Left	M53, M55
8340664	Fuel Cell Assembly	M59, M84
10918604	Tank, Fuel, Engine	M114, M114A1, M114A1E1, M114B2
10952618	Cell Assembly, Fuel, Engine (Center)	M551A1
11633639	Tank, Fuel, Engine	M132, XM806
11647693	Bladder, Fuel	M741

NOTE: Provisions will be made to include new drawing numbers in the above list when the drawings are added.

3.3.1 Carcass. The carcass shall be constructed by the hand-lay-up method, each laminar fully calendered and final combined layers fully calendered. There shall be no spray-on build-up permitted. Fabric reinforcements or coated fabrics may be used in the inner layer, provided the seams of the adjacent plies are not superimposed. If specified, a fuel resistant barrier may be placed between the inner layer and the outer portion of the carcass, to conform to 3.4.6 (see 4.7.2 and 6.2).

3.3.2 Access opening. Each fuel tank shall be provided with a hand hole and cover, as shown on applicable drawings, to permit arm access to the tank interior for inspection and repair (see 4.7.2).

3.3.3 Capacity The nominal capacity of the fuel tanks shall be as specified with a tolerance of ± 3.0 percent (%) (see 4.7.2 and 6.2).

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3.3.4 Threaded parts. Threaded parts shall conform to applicable specifications and drawings, and shall be in accordance with FED-STD-H28 (see 4.7.2).

3.3.5 **Installation**. The location of all fittings and the tank dimensions needed for installation shall be within the allowable tolerances specified on the applicable drawings (ace 4.7.3).

3.3.6 Construction.

3-3.6.1 Tank exterior (see 4.7.2).

3.3-6.1-1 Channels in liner laps. Channels in liner laps shall not exceed 1/8 by 3 inches nor be more than one in any linear foot of lap.

3.3.6.1-2 Channels between plies. Channels between plies at the buried edge of lap splices or in transition areas shall not exceed 1/8 inch.

3.3.6.1.3 **Channels around outer edge** of fitting flange. Channels around the entire outer edge of the fitting flange shall not exceed 1/8 inch in width.

3.3.6.1.4 **Channels around throat of fitting flange.** Channels around the entire throat of the fitting flange shall not exceed 1/8 inch in width.

3.3.6.1.5 **Delamination between plies**. Any delamination or ply separation between plies shall not exceed 1 inch. The average shall not exceed one delamination or ply separation per 5 square foot **(ft²)** area nor more than five in any one **5 ft²** area with not less than a 6 inch solid bond between delaminations.

3.3.6.2 Tank exterior (see 4.6.2).

3.3.6.2.1 **Channels or bridging of outer** plies Channels or bridging of outer plies at cord or fabric splice shall not exceed 1/4 inch in width of entire length of splice.

3.3.6.2.2 **Ply separation.** Any delamination or ply separation between any two plies, except liner and second ply, shall not exceed 1 inch. The average shall not exceed one delamination or ply separation per **5 ft²** area nor more than five in any one **5 ft²** area with not less than a 6 inch solid bond between delaminaions.

3.3.7 **Interchangeability**. All parts possessing the same functional and physical characteristics, equivalent in performance and durability, shall be directly and completely interchangeable without alteration of the part or adjoining items (see 4.7.2).

3.4 Performance The fuel tank shall be capable, when installed in the vehicle structure for which it is designed, of withstanding the loads incident to road shock and vibrations common to off-the-road and high speed highway operation of the vehicle.

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3.4.1 Leakage. The fuel tank shall show no evidence of leakage when filled with stoddard solvent (or equivalent), containing a staining agent, and subjected to an internal pressure of 1 1/2 times the maximum operating pressure of the intended system and allowed to stand for 30 minutes (see 4.7.4).

3.4.2 Handling. The fuel tank shall show no evidence of cracks, loosening of fittings, or failure of seams when it is folded and unfolded 40 times in accordance with standard procedure for shipment, and installed and removed five times (ace 4.7.5).

3.4.3 Slosh. The fuel tank shall show no evidence of leakage or other failure such as chafing, disengaged hangers, or liner breaks when filled, 2/3 full, with a fluid conforming to type III of TT-S-735 and subjected to rocking at approximately 15 degrees (o) angle from the vertical (300 total rocking angle) for 24 hours at e frequency of 16 to 20 cycles per minute (cpm), or for 40 hours at 10 to 16 cpm (see 4.7.6).

3.4.4 Stand. The fuel tank shall show no evidence of leakage when filled with a fluid conforming to type III of TT-S-735 and allowed to stand for 30 days at the standard ambient temperature (see 4.7.7)-

3.4.5 **Fluid resistance of exterior surfaces**. The exterior surfaces of the fuel tanks shall show no swelling, separation, blistering, or dissolution when immersed continuously, in a test fluid conforming to type III of TT-S-735, at standard ambient temperature for 24 hours (see 4.7.8).

3.4.6 Permeability. When fuel tank specimens are subjected to the test specified in 4.7.9, the permeability shall be less than 0.025 fluid ounce (fl-oz) per **ft²** per 24 hours for each specimen tested.

3.4.7 Fuel contamination. The nonvolatile material, for a sample of the inner layers of fuel tank, shall not exceed 60 milligrams (mg) per 100 milliliters (ml) of the contaminated fluid when the sample is diced into approximately 0.062 inch squares **(in²),** placed in a flask containing 230 ml of test fluid conforming to type 111 of TT-S-735, and allowed to stand for 48 hours at 77 ± 5 degrees Fahrenheit (°F). The stoved gum residue shall not exceed 20 mg per 100 ml of the contaminated fluid, after necessary corrections have been made for performed gums, originally present in the test fluid, and when the beakers, containing the nonvolatile material, are placed in an appropriate bath maintained constantly at a temperature of 572 ± 90°F for 30 minutes (ace 4.7.10).

3.4.8 Tensile strength. The inner layer ply of the tank, without barrier, shall have a tensile strength not less than 80% of the value calculated, based on the original cross sectional area of the ply, when subjected to test specified in 4.7.11.

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3.4.9 Seam adhesion. The seam adhesion of the inner ply of the fuel tank, where the adhesion of the seam is less than the strength of the material, shall be not less than 12 pounds (lb). There shall be no failure of the seam (see 4.7.12 or 4.7.12.1).

3.4.9.1 Wet seam adhesion. The wet seam adhesion shall be not less than 50% of the dry adhesion test value when subjected to the test specified in 4.7.12.2.

3.4.10 Puncture resistance. The force required to puncture the fuel tank wall shall be not leas than 15 lb when tested as specified in 4.7.13.

3.4.11 Low temperature leakage. The fuel tank shall show no evidence of leakage when filled with a test fluid conforming to type III of TT-S-735 and allowed to stand for a period of 7 days at 110 ± 10°F, emptied and subjected to air dry-out at 160 ± 10°F for 7 days, and then refilled with the same test fluid and cooled to a temperature of -67 ± 3.6°F for not less than 3 days (see 4.7.14).

3.4.12 Humidity (hydrolytic stability) at elevated temperatue. The specimen shall show no corrosion, peeling, cracking, warping, blistering, delamination, or discoloration when exposed to a temperature of 158 ± 2°F and 100% relative humidity for 15 days (see 4.7.15).

3.4.12.1 Humidity (hydrolytic stabililty) at room temperature. The specimen shall show no visible cracks, blisters or eruptions when examined with a 7-power magnifying glass after being exposed to room temperature and 100% relative humidity for 30 days (see 4.7.15.1).

3.5 Repair procedures. The repair procedures used to correct any discrepancies shall be only those methods approved by the Government (see 4.7.2).

3.6 Identification marking Identification marking on the fuel tanks shall be in accordance with MIL-STD-130. Each fuel tank shall have the following information permanently and legibly marked on the face or surface in which the opening for the fuel-level gage sending unit is located (see 4.7.2):

Army part number.
Vehicle nomenclature.
Name of manufacturer.
Date of manufacture (year and quarter).
Manufacturer-s serial number.
Manufacturer-a construction number.
Specification MIL-T-14442. .
Date of acceptance.
Contract or order number.

3.6.1 Filler cap. The external surface of the filler cap shall be formed and painted with diesel fuel (DF-A, DF-1, DF-2, and JP-8) resistant lacquer, and legibly marked in white lettering: “Fuel (*) US Gal.” (* to be filled in with usable capacity taken as the average capacity of the first 10 production tanks).

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3.6.2 Hand-hole cover The exterior surface of the hand-hole cover of each fuel tank shall be legibly and permanently marked "OUTSIDE".

3.6.3 Torque values. The torque values required for assembly shall be marked on, or adjacent to, each fitting or accessory involved, on the side torque wrench is applied. If the torque required is the came for each such item, that value may be stenciled in only one place on the surface of tank in a conspicuous location.

3.7 Workmanship. The workmanship shall be in accordance with high-grade manufacturing practice for the kind of material covered by this specification and shall be such as to produce fuel tanks free of the defects specified herein. Defects which may impair the serviceability of fuel tanks are detailed hereinafter, together with their specific limits, and measure to be taken for correction, when permissible (see 4.7.2).

3.7.1 Tank interior (see 4.7.2).

3.791.1 Blisters between liner laps. Blisters between inner liner laps shall not exceed 1/4 inch nor average more than one per 5 linear feet of splice, with a maximum of five in any one 5 feet length of splice.

3.7.1.2 Blisters between plies. Blisters between plies shall not exceed 1/4 inch and shall be not less than 6 inches between blisters nor be more than one per ft^2 of tank area.

3.7-1.3 Blisters between fitting flange and adjacent ply. Blisters between fitting flange and adjacent ply shall not exceed 1/4 inch nor be more than one in any one linear foot of flange area nor more than one per fitting when flange is leas than one linear foot.

3.7.1.4 Blisters between cemented components Blisters between cemented components and liner, or other components, including chafing strips, shall not exceed 15% looseness and shall have a minimum of 1/4 inch continuous bond around the edge.

3.7.1.5 Gum inner liner fried or scarred Fried or scarred portions of gum inner liner shall not exceed one-half of specified thickness of liner.

3.7.1.6 Cuts or holes. The tank interior shall be free of cuts or holes in the inner liner material.

3.7.1-7 Damaged inner liner material. Inner liner material damaged from mishandling shall not excee one-half of specified liner thickness.

3.7.1.8 Foreign material. Foreign material in the inner liner material shall not exceed one-half of specified liner thickness.

3.7.1.9 Thinned inner liner caused by buffing. Thinned inner liner material caused by buffing shall not exceed one-third of specified liner thickness.

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3.7.1.10 Imperfect inner liner material. Imperfect or missing lamination of inner liner material shall not exceed 1/4 inch in diameter nor be more than 1/3 of specified liner thickness.

3.7.1.11 Inner liner defects causing exposed fabric. The inner liner material shall be free of any defects causing fabric of the fuel tank to be exposed.

3.7.1.12 Inner liner damage causing exposed fabric inner liner. There shall be no indications of inner liner damage caused by buffing through fabric inner liner.

3.7.1.13 Inner liner skin coat blister. There shall be no indication of inner liner coat blisters.

3.7.1.14 Loose seams in inner liner laps. The tank interior shall be free of loose seams in the inner liner laps that cannot be repaired by trimming.

3.7.1.15 Loose cemented components. Loose cemented components, such as chafing strips, reinforcement straps, or tabs, shall not exceed 15% of total area.

3.7.1.16 Loose liner reinforcements. There shall be no indication of loose liner reinforcements that cannot be repaired by triming.

3.7.2 Tank exterior (see 4.7.2).

3.7.2.1 Damaged cord or fabric ply. The exterior of the tank shall be free from damage through any single cord or fabric ply-

3.7.2.2 Void around throat of fiting. Void completely around throat of fitting shall not exceed 1/8 inch in width:

3.7.2.3 Lap splice edge looseness. Lap splice edge looseness not exceeding 1/8 inch in depth nor 3 inches in length is acceptable, provided repair can be made by trimming, and provided no more than one such defect occurs per 5 linear feet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchaee order (see 6.2), the contractor is responsible for the performancee of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor-s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance shall comply with all requirements of the contract. sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformancee to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article Inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions Unless otherwise specified (ace 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 3.6°F
b. Barometric pressure: 26.5 ± 2 inches mercury
   - 3
c. Relative humidity: 50 ± 5%

4.3.1 Test structure. When specified (see 6.2), the teat structure shall be a simulation of the actual vehicle structure. It shall duplicate the shape, dimensions and material of the tank supporting structure in the vehicle. The necessary stops, cushions, and pads, identical with those used in the finished vehicle, for mounting and supporting the tank, shall be provided. It shall be constructed so that the test tank can be installed in and removed from the structure in an identical manner to that used in the actual vehicle installation.

4.3.2 Support jig. The jig shall support the tank test structure specified in 4.3.1, to simulate the actual vehicle installation. The support jig shall be a strongly constructed framework of metal, for carrying the mounted test tank and designed for bolting to the rocker assembly specified in 4.7.6. The jig shall be designed so that removal of the tank test structure may be accomplished with a minimum effort.

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4.4 First article inspection Unless otherwise specified (see 6.2), the Government shall select the following tank samples produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply fuel tanks that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from

officer. First article samples shall be selected as follows:

Three full size tanks complete with necessary fittings-
Three 12 by 12 inch specimens of composite tank construction.
One specimen of inner layer ply, without barrier, approximately 900 in^2 in area, including seam.
One specimen of 6 by 6 inch inner layer ply, without barrier.
Three permeability specimens in accordance with 4.7.9.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Test	
Materials and construction	3.2 thru 3.3	4.7.1	X			
Defects (see 4.5.2 and table III)	3.3 thru 3.3.4, 3.3.6, 3.3.7, and 3.5 thru 3.7.2.3	4.7.2	X	X		
Installation	3.3.5	4.7.3.1 4.7.3.2	X		X	
Leakage	3.4.1	4.7.4	X			
Handling	3.4.2	4.7.5	X			
Slosh	3.4.3	4.7.6	X			
Stand	3.4.4	4.7.7.1 4.7.7.2	X			X
Fluid resistance of exterior surfaces	3.4.5	4.7.8	X			
Permeability	3.4.6	4.7.9	X			
Fuel contamination	3.4.7	4.7.10	X			
Tensile strength	3.4.8	4.7.11	X			
Seam adhesion	3.4.9	4.7.12 & 4.7.12.1	X			

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Test	Control
Wet seam adhesion	3.4.9.1	4.7.12.2	X			
Puncture resistance	3.4.10	4.7.13	X			
Low temperature leakage	3.4.11	4.7.14	X			
Humidity at elevated temperature	3.4.12	4.7.15	X			
Humidity at room temperature	3.4.12.1	4.7.15.1	X			

4.4.1 First article inspection failure. Deficiencies found during, or first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article inspection, shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 Quality Conformance inspection (QCI). QCI shall include the examination of 4.5.2 and the tests of 4.5.3. 'Noncompliance with any of the specified requirements in sections 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.5 for definitions of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all the fuel tanks of one part number, from an identifiable production period, from one manufacturer, from one manufacturing location, submitted at the same time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table II.

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TABLE II. Sampling plan for QCI.

Inspection lot size	Sample size		
	Examination		
	Major	Minor	Test
2 to 8	*	5	*
9 to 15	13	5	13
16 to 25	13	5	13
26 to 50	13	5	13
51 to 90	13	7	13
91 to 150	13	11	13
151 to 280	20	13	20
281 to 500	29	16	29
501 to 1200	34	19	34
1201 to 3200	42	23	42
3,201 to 10,000	50	29	50
10,001 to 35,000	60	35	60
35,001 to 150,000	74	40	74
150,001 to 500,000	90	40	90
500,001 and over	102	40	102

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table III (see 4.7.2). The acceptance number in all cases is zero.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Improper materials (see 3.2).	Visual
102	Assembly incomplete (see 3.3 thru 3.3.7).	Visual
103	Dimensions affecting interchangeability, out of tolerance (see 3.3 thru 3.3.7).	SIE 2/
104	Improper marking (see 3.6).	Visual
105	Faulty workmanship (see 3.7 thru 3.7.2.3). 1/	Visual and functional

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TABLE III. Classification of defects - Continued.

Category	Defect	Method of examination
Minor		
201	Improper materials, protective coating and coverage (see 3.2).	Visual
202	Dimensions not affecting interchangeability, out of tolerance (see 3.3 thru 3.3.7).	SIE 2/
203	Improper marking of data and instructions (see 3.6).	Visual

1/ A 1-inch minimum total effective bond shall be maintained between all blisters and in all laps, except in construction where specific approval has been granted for a lesser minimum total effective bond. In such cases, the approved minimum applies.
2/ SIE = Standard Inspection Equipment.

4.5.3 Tests. The sample selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table I. The acceptance number in all cases is zero.

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected and resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection iota may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.6 Control tests. Control test(s) shall be conducted on one fuel tank out of the first lot of 50 units consecutively produced, except that not more than one test shall be performed in a one month period, nor less than one test in a two month period. The fuel tank assemblies shall be subjected to the control tests specified in table I.

4.6.1 Failure. Failure of any fuel tank to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.7 Methods of inspection.

4.7.1 Materials, design and construction. Conformance to 3.2 through 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications. design data, receiving inspect ion records, processing end quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3 through 3.3.4, 3.3.6, 3.3.7, and 3.5 through 3.7.2.3 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7.3 Installation.

4.7.3.1 First article sample. To determine conformance to 3.3.5, the tank shall be checked in the test structure specified in 4.3.1 for the location of all fittings and for the tank dimensions.

4.7.3.2 Quality conformance. To determine conformance to 3.3.5, the tank shall be checked in a test jig or registered metal templates fabricated in accordance with design requirements furnished by the vehicle manufacturer for the particular vehicle (see 4.3.2). The location of all fittings and the tank dimensions shall be checked. Nonconformance of one or more samples shall be cause for rejection of the lot.

4.7.4 leakage. To determine conformance to 3.4.1, the tank shall be covered with brown paper, filled with a stoddard solvent, and supported systems shall be pressurized and allowed to stand for a period of time as specified. Tanks intended for use in non-pressurized systems shall be subjected to an internal pressure equivalent to the normal head pressure measured at the bottom of the tank and multiplied by a factor of 1.5 and allowed to stand for 30 minutes. The brown paper shall then be examined for stains.

4.7.5 Handling To determine conformancee to 3.4.2, prior to installation in the-test structure (see 4.3.1), the fuel tank shall be folded and unfolded as specified and then installed and removed from the teat structure five times. Applicable service procedures shall be followed in tank installation and removal . All fittings shall be fastened to corresponding structure and interconnect fittings upon each installation in the test structure, and the tank shall then be examined.

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4.7.6 Slosh. To determine conformance to 3.4.3, the tank shall be weighed to the nearest ounce and the weight shall be recorded. The test tank, complete with all fittings and other parts or accessories, shall be mounted in the support jig (see 4.3.2) and installed on a rocker assembly. It shall be mounted in such a manner that the major (longer) horizontal axis of the tank shall be 90 degrees (°) to the center line of the axis of the rocker assembly platform. The interior of the compartment shall be lined with brown paper in the fitting areas and held in place with tape conforming to PPP-T-76. The test specimen shall be filled two-thirds full with test fluid as specified. In addition, twice the sump capacity of water shall be added to the tank. The tank shall then be subjected to rocking at a frequency, period of time, and rocking angle as specified. The temperature of the test fluid during the test shall be 110 ± 10°F. Tanks intended for use in pressurized fuel systems shall be subjected to an internal pressure of 1-1/2 times the maximum operating pressure of the intended vehicle system during the entire test. Tanks intended for use in non-pressurized fuel systems shall be tested at normal atmospheric pressure. When testing non-pressurized tanks, a vent opening shall be provided equal to the vent opening in the intended vehicle. The vent shall be connected to a line and so positioned that any test fluid escaping through the vent shall be carried away from the test specimen. At the conclusion of the 24 or 40 hours rocking period, as applicable, the tank shall be completely filled with the specified test fluid and allowed to stand for 1 hour. The test fluid shall then be drained and the test tank removed from the structure. The paper lining shall be thoroughly examined for leakage as indicated by stains on the paper. Indication of stains on the liner paper shall be considered as test failure, except minor stainage due to improperly torqued bolts, damaged equipment not considered a part of the tank assembly, or minor stainage resulting from physical damage to the tank at the time of installation.

4.7.7 Stand. To determine conformance to 3.4.4, the following shall be examined:

4.7.7.1 First article. Upon completion of the low temperature test specified in 4.7.14, the cavity shall be completely lined with brown paper and the tank shall be emptied and refilled with a test fluid conforming to type 111 of TT-S-735. The tank shall be allowed to stand at the standard ambient temperature for a period of 30 days, at which time the paper shall be examined for any stains indicating a fluid leakage.

4.7.7.2 Periodic control samples. Bach tank selected in accordance with 4.6 shall be collapsed and held for 30 minutes in a position comparable to that encountered prior to installation in its respective vehicle cavity. Prior to the installation of the tank, the test structure shall be completely lined with brown paper and held in place with tape conforming to PPP-T-76. Then the tank shall be installed in its test structure. The tank shall then be filled with a test fluid conforming to type III of TT-S-735, and stand tested for 30 days. The tank may be inspected at 10-day intervals during the

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test. At the conclusion of the 30-day test, the tank shall be drained and removed from the test cavity. The test cavity shall be inspected for evidence of leakage as indicated by stains on the brown paper. After the examination, if stains are present on the brown paper, evaluation shall be made in accordance with the evaluation procedure specified in 4.7.6. If staining occurs that cannot be attributed to one of these causes, the cause of leakage shall be determined by Government inspection, contractor inspection, engineering and laboratory personnel. Their determination shall include dissection of the teat specimen at strategic points, if necessary, to determine cause of failure.

4.7.8 Fluid resistance of exterior surfaces. To determine conformance to 3.4.5, a 6 by 6 inch specimen of composite tank construction shall be submerged in a container of test fluid at standard ambient temperature for a period as specified. Specimen shall be removed after continuous immersion and examined.

4.7.9 Permeability. To determine conformance to 3.4.6, the fuel tank shall be tested as specified in 4.7.9.1 through 4.7.9.3.

4-7-9-1 Test apparatus. The test apparatus shall include a permeability cup and ring constructed in accordance with figure 1. A nylon solution shall be used for sealing the test disk to permeability cup.

4.7.9.2 Preparation of test specimen. The uncured liner shall be applied to a 10 by 10 inch piece of corrugated fiberboard coated on one side with a suitable water soluble breakaway agent. The exposed surface of the inner liner shall be coated with prime cement and barrier resin, if required, iin accordance with applicable manufacturing specifications. The assembly shall be wrapped with cellophane and covered with a waterproof bag. The assembly shall be vulcanized using the same method as in regular production. After vulcanization, the waterproof bag and cellophane shall be removed, and the inner liner shall be removed from the fiberboard, using water if necessary. Free moisture shall be wiped from the assembly, and the assembly shall be conditioned for 24 hours at a constant temperature of 77 ± 50°F and a relative humidity of 40 ± 5%. Two disks, each 2.5 inches in size, shall be cut from the vulcanized panel. 100 ml of the test fluid conforming to type III of TT-S-735 shall be placed in the permeability cup shown in figure 1. A nylon solution shall be applied to the face of the cup flange covering the area inside the bolt circle. The nylon solution shall be allowed to come almost to dryness, then the test disk shall be applied to the cup with the barrier, if any, facing outward. The assembly shall be completed by attaching the bolting ring and tightening the bolts in accordance with the following schedule:

Inner linerr type	Bolt torque, pound-inches (lb-in)
Gum stock	5 to 10
Coated fabrics	15 to 20
Unsupported plastic films	20 to 25

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4.7.9.3 Method of conducting test. Permeability CUPS. prepared as specified in 4.7.9.2, shall be placed in a rack in a constant temperature of 77 ± 5°F and relative humidity of 40 ± 5%. After allowing one hour for equilibration, the CUP shall be weighed to the nearest 0.005 gram (g) and placed in the rack with the cup facing upward. The cup shall be kept at the above constant temperature for 24 hours, then weighed to check for excessive vapor loss. The bolts shall be re-torqued if necessary. The cup shall be inverted (test disk down) in a rack that permits free access of air to the test disk. Cups shall be weighed at the end of the third, fifth. and eight day after inverting. Defective films or leaks caused by faulty assembly will usually be found when making the weighing on the third day. The diffusion rate calculation shall be made on the fifth to the eighth day period, and the permeability shall then be determined. Diffusion expressed in fl-oz per ft^2 per 24 hours equals the gram loss of the test specimen per 24 hour multiplied by a factor K which is defined as follows:

$$K = \frac{144}{(\text{sp. gr.})\ (29.573)\ (3.1416)\ (R)^2}$$

Where:

Sp. gr. = specific gravity of the test fluid at 77°F.
R = inside radius of test cup expressed in inches.

4.7.10 Fuel contamination To determine conformance to 3.4.7, the fuel tank shall be tested as specified in 4.7.10-1 and 4.7.10.2.

4.7.10-1 Nonvolatile gum residue. A 5 gram sample of the inner layers, up to the barrier, shall be diced and placed in a flask containing a test fluid as specified, and allowed to stand for 48 hours at 77 ± 5°F. The contaminated test fluid shall be decanted off, and the nonvolatile gum residue determined by the method specified in ASTM D381, the total evaporation time shall be 45 minutes.

4.7.10.2 Stoved gum residue. The beakers containing the nonvolatile material shall be placed in an appropriate bath as specified, and then weighed after being cooled in a closed container.

4-7-11 Tensile strength. To determine conformance to 3.4.8, the tensile strength of a sample of inner layer ply, without barrier, shall be determined in accordance with the test specified in ASTM D412, when coated fabric liners are used. The sample shall then be immersed in water at 180 ± 4°F for 72 hours, after which the specified tensile strength test shall be repeated. The change in tensile strength shall be reported as the after-immersion strength, calculated as a percentage of the *original* value.

4.7.12 Seam adhesion. To determine conformance to 3.4.9, the seam adhesion of the inner layer ply to itself shall be tested along the length of the seam in accordance with the test specified in ASTM D413, using the strip specimen.

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4.7.12.1 Seam adhesion (alternative procedure). To determine conformance to 3.4.9, the seam adhesion of the inner layer ply to itself may be tested by cutting a strip of inner-layer material one inch wide, having a seam made in the same manner as is used in the tanks submitted under 4.4. This seam shall be perpendicular to, and midway in, the length of the strip. A tensile load shall be applied parallel to the length of the strip, of sufficient magnitude to break the strip.

4.7.12.2 Wet seam adhesion. To determine conformance to 3-4.9.1, a specimen shall be prepared in accordance with the test of 4.7.12 or 4.7.12.1, and soaked in a test fluid conforming to type III of TT-S-735 for a period of 48 hours, at 75 ± 5°F. The specimen shall be removed from the test fluid and wiped dry. No other means of drying shall be used. Within 4 hours after removing the specimen from the fluid, it shall be subjected to one of the above tests. The value by the strip method shall be no less than 50% of the dry test value. The results using the alternative procedure shall require the breaking of the strip outside of seam area-

4.7.13 Puncture resistance. To determine conformance to 3.4.10, a test specimen of tank wall construction shall be fastened in a specimen holder as shown in figure 2. A piercing instrument, with its piercing end as shown in figure 2, shall be forced against the specimen in the holder at approximately the center of the area enclosed by the specimen holder. The force required to puncture the specimen shall be not less than specified.

4.7.14 Low temperature leakage. To determine conformance to 3.4.11, the fuel tank shall be emptied, after the test specified in 4.7.6, and the interior of the cavity in which the tank is installed shall be completely lined with brown paper. The tank shall be refilled with a test fluid as specified, containing a staining agent, and allowed to stand for a period of 7 days at 110 ± 10°F. The tank shall then be emptied end subjected to an air dry-out as specified. Following the air dry-out, the tank shall be refilled with the same test fluid specified and cooled to a temperature as specified and allowed to remain at this temperature for a minimum period of 3 days. The tank shall then be brought back to room temperature and allowed to stand until completely drained and then examined for leakage as indicated by staining the brown paper.

4.7.15 Humidity (hydrolytic stability) at elevated temperature. To determine conformance to 3.4.12, one 6 by 6 inch *specimen* of the composite tank construction shall be examined after being exposed to a temperature and a relative humidity, as specified, for a period of 15 days.

4.7.15.1 Humidity (hydrolytic stability) at room temperature. To determine conformance to 3.4.12-1, two specimens 1 by 6 by .075 ± .025 inches shall be examined after being exposed to room temperature and relative humidity, as specified, for 30 days. *When* practical, these specimens shall be mounted on an appropriate fixture in a manner to cause a stretch of 20%. measured between gage marks 4 inches apart, centered in the middle section of each strip.

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5. PACKAGING

5.1 **Preservation, packaging, packing, and marking.** Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements as specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nonmetallic, non-self-sealti fuel tanks covered by this specification are intended for installation unmilitary tactical vehicles including trucks, truck-tractors, and similar thin-skinned wheeled vehicles, as reservoirs for compression ignition fuel conforming to type III of TT-S-735. They may also be installed on combat vehicles, to provide corrosion-resistant fuel storage reservoirs. The fuel tanks are intended for use with fuel which does not exceed 30 percent aromatics content.

6.2 Acquistion requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Whether first article inspection is rewired (see 3.1).
d. Title, number, and date of applicable drawings (see 3.3).
e. Nomenclature of vehicle or vehicles in which the fuel tanks are intended to be installed (see 3.3).
f. If a fuel resistant barrier is required in the carcass (see 3.3.1)-
g. Capacity of the fuel tank (see 3.3.3).
h. If responsibility for inspection shall be other than as specified (see 4.1).
i. If inspection conditions shall be other than as specified (see 4.3).
j. A test structure conforming to 4.3.1 to support first article inspection, when applicable.
k. Arrangements for first article inspection and rights of the Government (see 4.4 and 6.3).
l. If sampling plan for QCI is other than as specified (see 4.5.1).
m. Selection of applicable level and packaging requirements (see 5.1).

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(e) should be a first article sample, a first production item, or a standard production item from the contractor-s current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the

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Government reserves the right to waive the requirement for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Rubber material**. Satisfactory results have been obtained with fuel tanks constructed with fabric-reinforced nitrile rubber.

6.5 Definitions.

6.5.1 **Definition of terms used with sampling inspection**. The following terms are used in relation with sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or *more* of the following classes: critical, major and minor defects. Also , defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. **Critical defective**. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. **Defective**. A defective is a unit of product which contains one or more defects.

e. **Formation of lots or batches**. The product shall be assembled into identifiable iota, sublets, batches, or in such other *manner as may be* prescribed (see 6.5.1.1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "'inspection lot" or "inspection batch"', i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

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g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublets or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample. A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

P. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

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6.5.2 **Recovered materials** Recovered materials means materials that have been collected or recovered from solid waste (see 6-5.3).

6.5.3 Solid waste. l 'Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material; including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6 **Subject term (key word) listing.**

Access opening
carcass
Configurations
Handling
Installation
Interchangeability
Leakage
Slosh
Tensile strength
Workmanship

6.7 **AMC policy on AQLs/LTPDs.** This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defective) from military specifications-

6.8 **Changes from previous issue.** Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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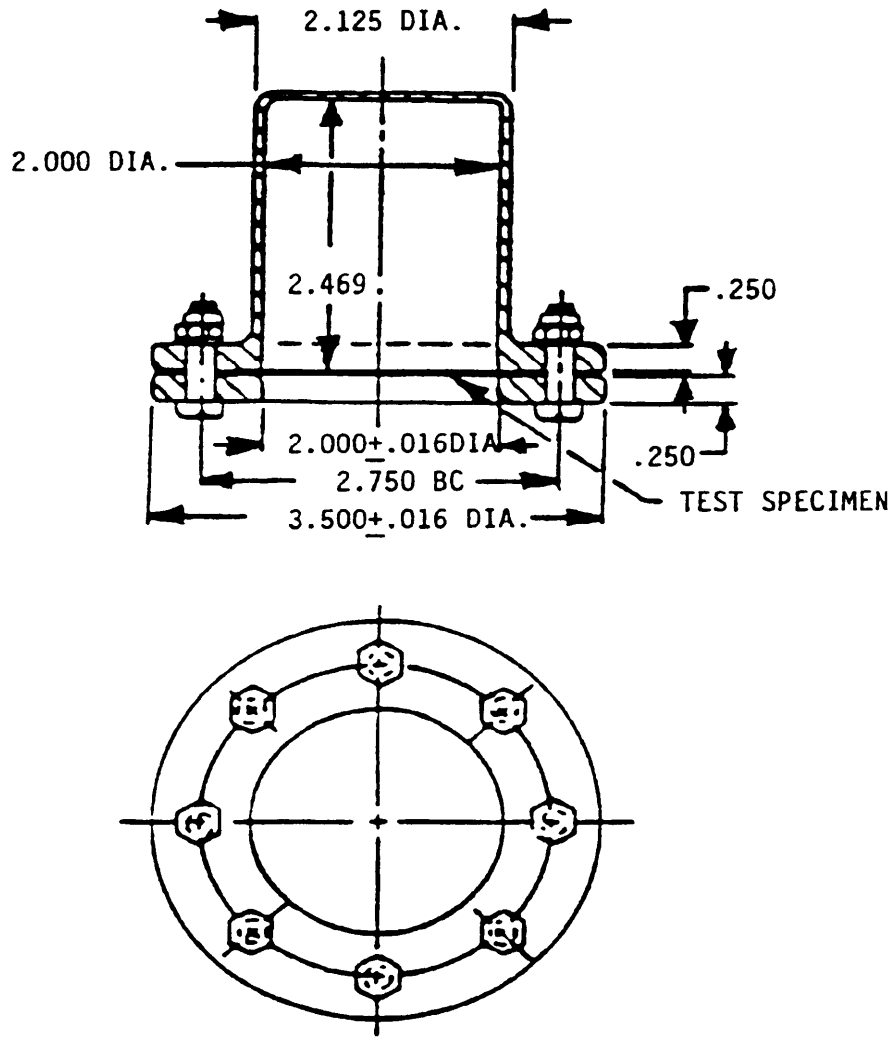


FIGURE 1. Diffusion cup assembly.

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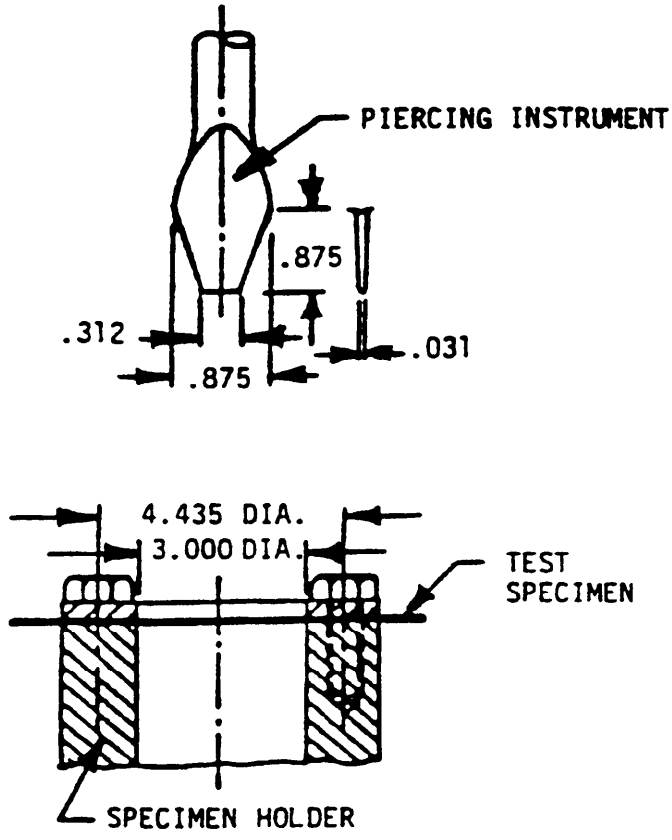


FIGURE 2. Piercing instrument and specimen holder.

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Preparing activity:
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Review activities:
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